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VLM Council — Evaluation Report

COUNTRY ACCURACY

64.0%

MEDIAN HAVERSINE

439 km

IMAGES

500

PATH A / PATH B

315 / 185

REGION NARROWING QUALITY

0.71

HYPOTHESIS POOL QUALITY

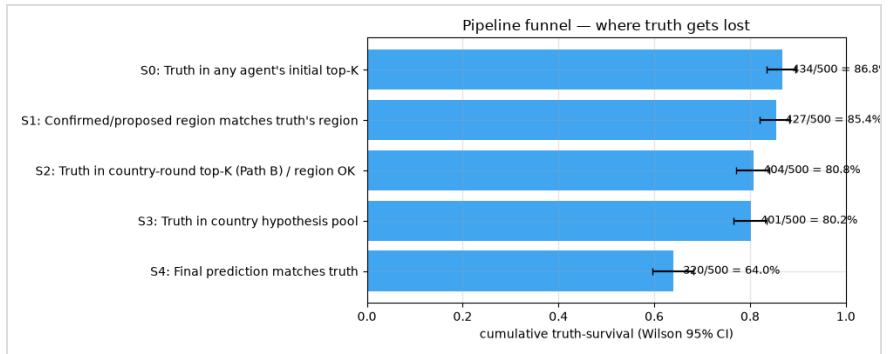
0.43

Pipeline Funnel

Where does the truth country get lost in the Progressive Narrowing pipeline?

Stage	Description	Cumul. survival	Conditional
S0	Truth in any agent's initial top-K	86.8% (434/500)	86.8%
S1	Confirmed/proposed region matches truth's region	85.4% (427/500)	98.4%
S2	Truth in country-round top-K (Path B) / region OK (Path A)	80.8% (404/500)	94.6%
S3	Truth in country hypothesis pool	80.2% (401/500)	99.3%
S4	Final prediction matches truth	64.0% (320/500)	79.8%

Bottleneck: S4 — Final prediction matches truth
Conditional rate: **79.8%** (95% CI [75.6%, 83.4%])

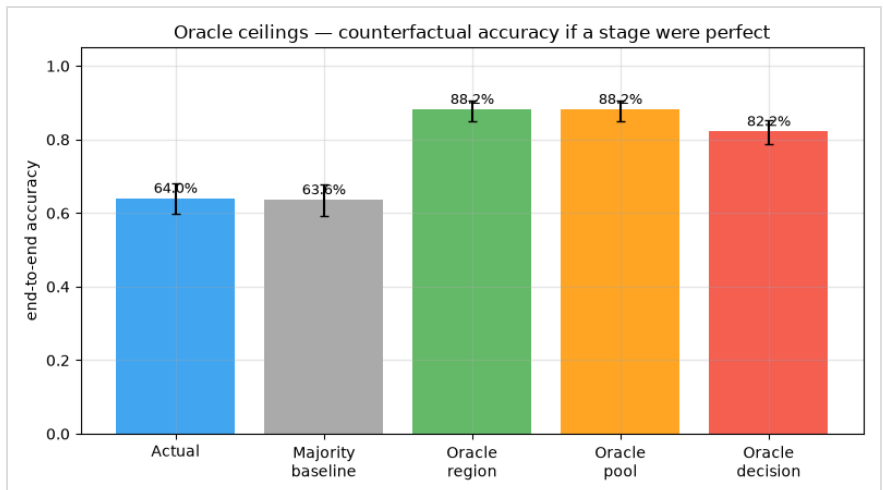


Cumulative truth-survival through the PN pipeline (Wilson 95% CI).

Oracle Ceilings

Counterfactual accuracy if a given stage were perfect.

Scenario	Accuracy	n
Actual	64.0%	500
Majority-vote baseline (initial round)	63.6%	500
Oracle region — perfect region step	88.2%	500
Oracle pool — truth always in hypothesis pool	88.2%	500
Oracle decision — perfect final judge	82.2%	500

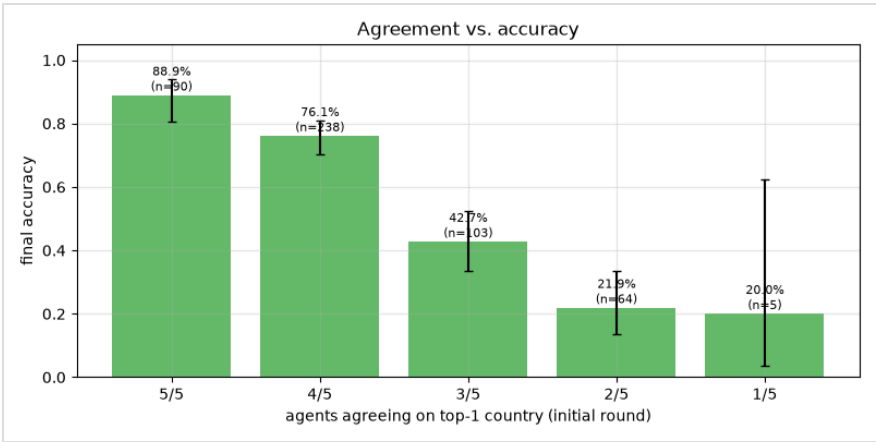


Oracle ceiling bars — gap between actual and each ceiling is the recoverable headroom.

Agreement vs. Accuracy

Final accuracy split by how many of the 5 agents agreed on the same top-1 country in the initial round.

Agents agree	n	Accuracy	CI low	CI high
5/5	90	88.9%	80.7%	93.9%
4/5	238	76.1%	70.2%	81.0%
3/5	103	42.7%	33.6%	52.4%
2/5	64	21.9%	13.5%	33.4%
1/5	5	20.0%	3.6%	62.4%



Higher initial agreement correlates strongly with final accuracy.

Path A vs. Path B

Metric	Path A (consensus)	Path B (no consensus)
n	315	185
Accuracy	73.0%	48.6%
Median haversine	323 km	936 km

Metric	Path A (consensus)	Path B (no consensus)
Truth in hypothesis pool	91.4%	66.5%

Error Severity

NEAR MISS (< 500 KM)

41

SAME REGION, WRONG
COUNTRY

114

WRONG REGION

58

► Near-miss examples (41)

► Wrong-region examples (58)

Geo-spatial bias

- North/south: strong north bias ($p=0.0067$)
- East/west: no significant bias ($p=0.8476$)

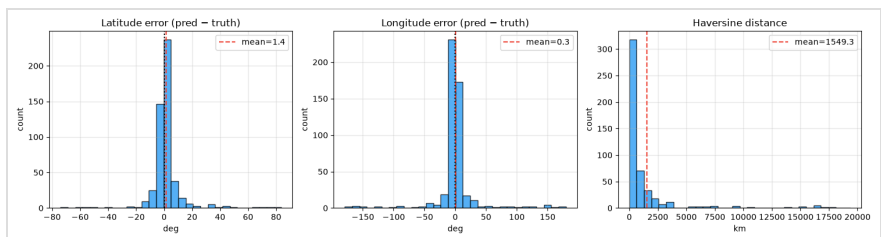


Fig 1: Lat / lng / haversine error distributions.

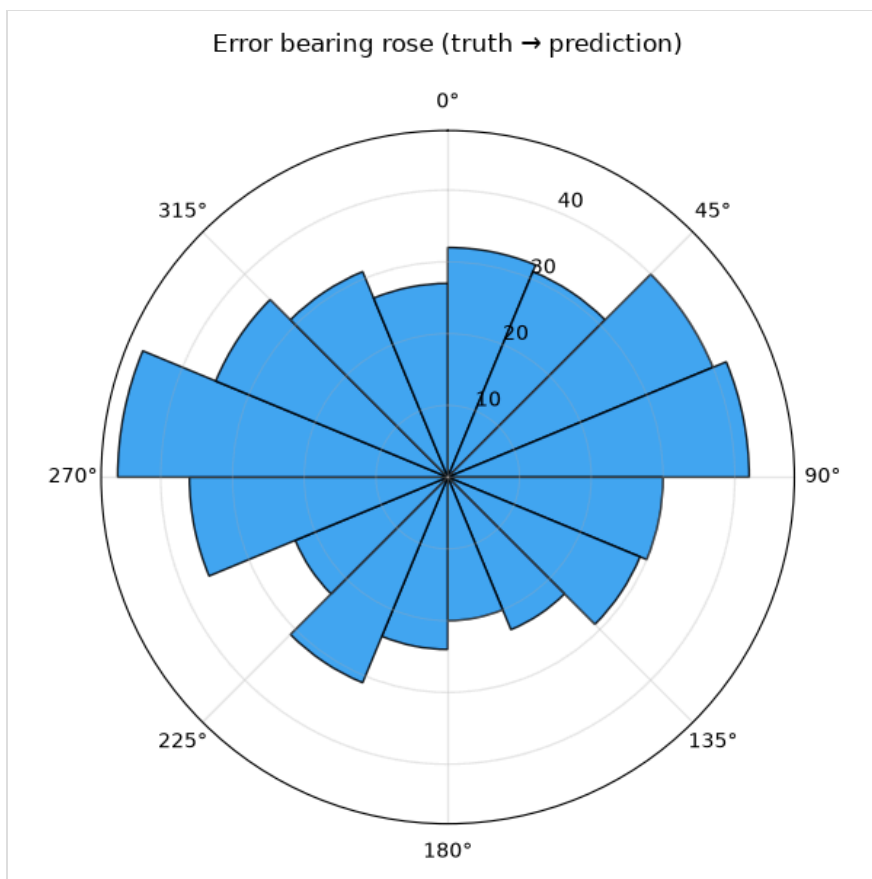


Fig 2: Direction of prediction errors.

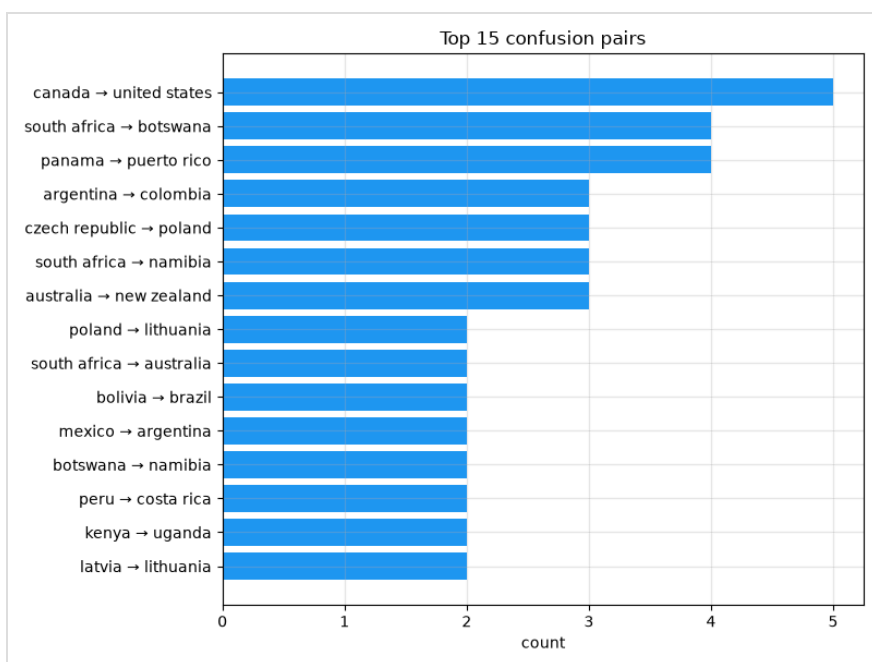


Fig 3: Top confusion pairs (truth → predicted).

► Top confusion pairs (20)

► Asymmetric confusion pairs ($A \rightarrow B \neq B \rightarrow A$)

Per-agent metrics

Initial round

Agent	n	Top-1	Top-3	Coverage
linguistic	107	78.5%	86.0%	91.6%
landscape	500	63.4%	79.8%	81.6%
botanics	499	63.1%	78.6%	79.6%
regulatory	443	64.6%	77.7%	78.6%
meta	499	62.5%	74.7%	74.7%

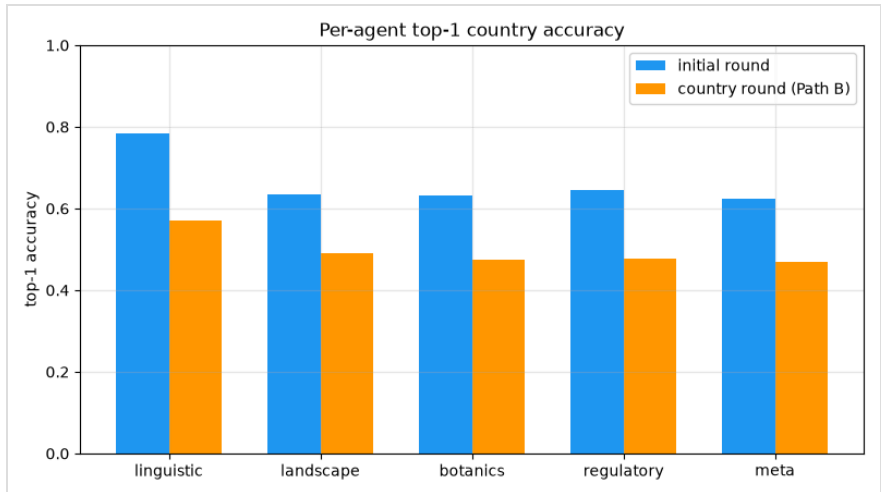


Fig 4: Per-agent top-1 accuracy (initial vs. Path-B country round).

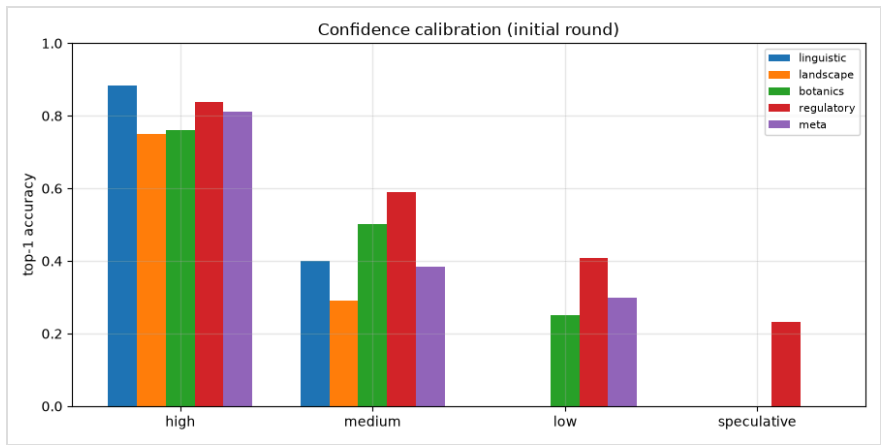


Fig 5: Confidence calibration by agent.

Country-round — Path B only

Agent	n	Top-1	Top-3	Coverage
linguistic	35	57.1%	71.4%	74.3%
landscape	185	49.2%	61.6%	62.2%
botanics	185	47.6%	60.0%	60.5%
regulatory	184	47.8%	60.3%	61.4%
meta	185	47.0%	55.1%	55.1%

Progressive Narrowing scores (LLM judge)

Verdicts: **500** / 500 · Constructive synthesis: **69.4%**

Region narrowing quality

How well-reasoned was the region decision? For Path A (consensus) this scores the consensus check; for Path B it scores the region-decision reasoning quality.

Split	n	Mean	Median	Stdev
All	500	0.709	0.750	0.312
Path A (consensus)	0	—	—	—

Split	n	Mean	Median	Stdev
Path B (no consensus)	0	—	—	—

Hypothesis pool quality

Were the generated country hypotheses the right set of plausible candidates for the image? Scored regardless of whether the truth country was in the pool.

Split	n	Mean	Median	Stdev
All	500	0.435	0.500	0.279
Truth was in pool	0	—	—	—
Truth NOT in pool	500	0.435	0.500	0.279

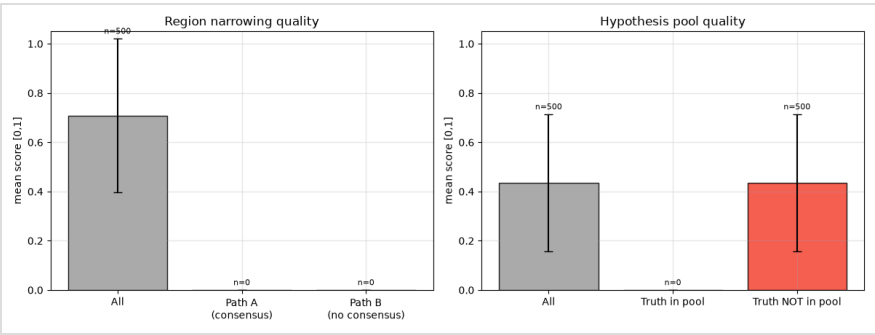


Fig 6: Region narrowing quality (Path A vs B) and hypothesis pool quality.

Agent judge verdicts

Agent	n	Role adh.	Halluc. ↓	Visual cons. ↑	Conf. calib. ↑
linguistic	500	100.0%	0.00 ± 0.00	0.89 ± 0.32	0.49 ± 0.28
landscape	500	100.0%	0.02 ± 0.06	0.93 ± 0.07	0.56 ± 0.34

Agent	n	Role adh.	Halluc. ↓	Visual cons. ↑	Conf. calib. ↑
botanics	500	100.0%	0.03 ± 0.10	0.90 ± 0.10	0.54 ± 0.32
regulatory	500	100.0%	0.01 ± 0.03	0.94 ± 0.07	0.55 ± 0.30
meta	500	100.0%	0.02 ± 0.08	0.90 ± 0.10	0.55 ± 0.34

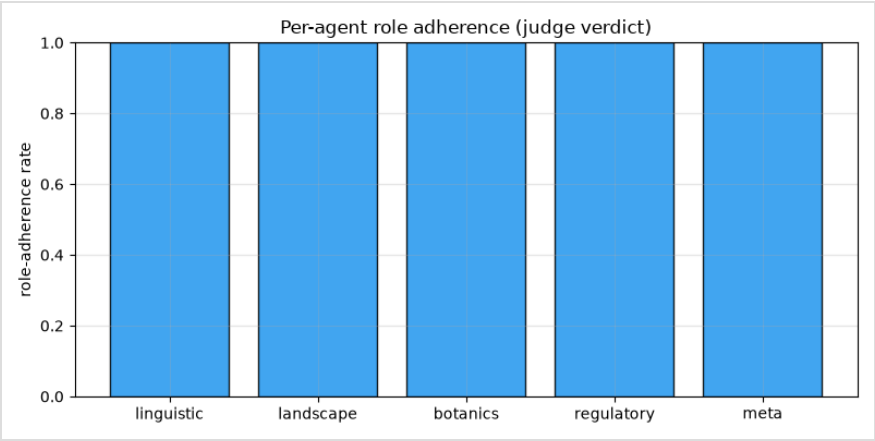


Fig 7: Per-agent role adherence rate.

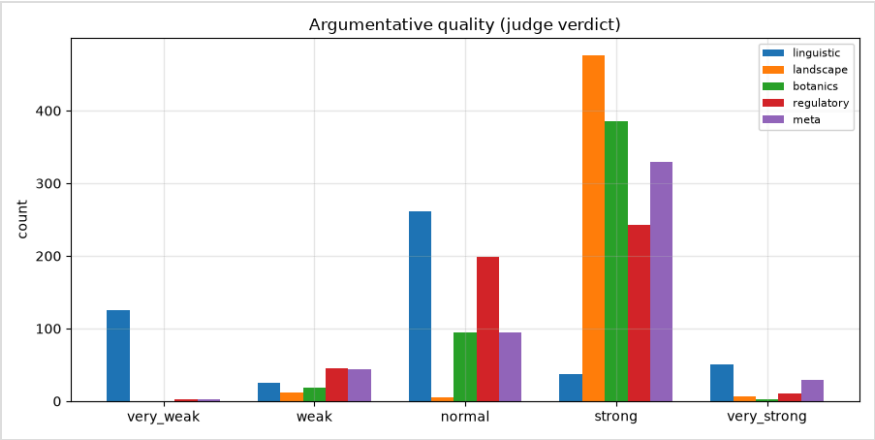


Fig 8: Argumentative quality histogram.

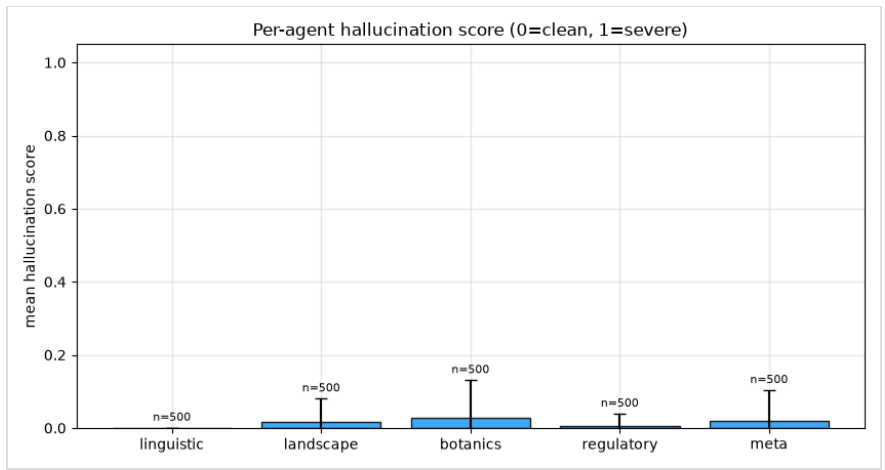


Fig 9: Per-agent hallucination score (0 = clean, 1 = severe).

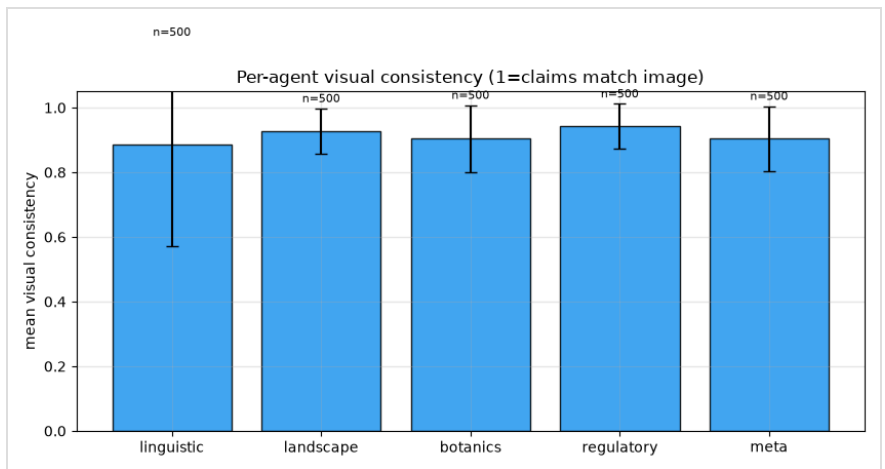


Fig 10: Per-agent visual consistency.

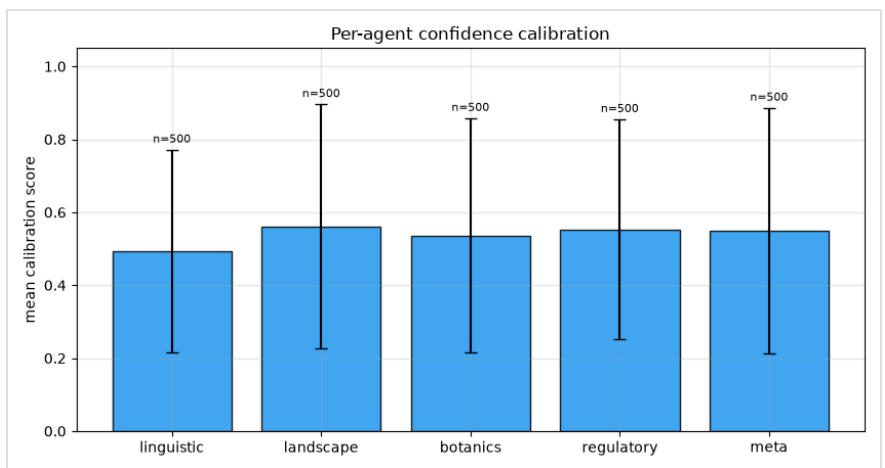


Fig 11: Per-agent confidence calibration.

Hallucination examples

▶ landscape — 5 flagged claim(s)

▶ botanics — 5 flagged claim(s)

▶ regulatory — 5 flagged claim(s)

▶ meta — 5 flagged claim(s)